Round 1 - Computer Vision Engineer - Aman Kumar

Round 1 - Online assessment for Computer Vision Engineer role at FOG.

Please **copy this document**, replace your **name in the document title**, fill in the candidate details below, answer all questions, and **submit your completed document via the provided Google Form**.

Candidate Name: Aman Kumar
Form Link: https://forms.gle/3VmdwqJHsoVZ6GqJ8

Instructions -

- The assessment should be done solely by yourself.
- Submit the response to assessment within 3 days.

Tips to succeed -

- Pay attention to detail
- Think user first
- Be creative
- Use AI (Chat GPT, Midjourney, Claude, v0, etc...)

All the best and don't forget to have fun!

Candidate Name: Aman Kumar
Website : https://scoreboard-extraction-uay8nkl5ynedyezwvh2pjg.streamlit.app/
GitHub repository URL : https://github.com/aman15cs21/scoreboard-extraction.git
Demo Video :

https://drive.google.com/file/d/148r4DlFA3ogxupMYfTWeNV4dtX1n0_-o/view?usp=sharing

Question 1: Scoreboard Data Extraction from Video

Develop a Computer Vision solution to **extract scoreboard data from a video**.

The system should process the video and extract the scoreboard information accurately from the frames.

Submission Format

1. GitHub Repository

Submit the **GitHub repository URL** containing the complete source code and a `README.md` file with instructions to run the project.

bowling_scoreboard.mp4

GitHub repository URL : https://github.com/aman15cs21/scoreboard-extraction.git

2. Demo Video

Submit **one video demonstrating the complete working solution**.

The video should show:

- Input video

- Code/project running
- Scoreboard being detected/extracted
- Extracted scoreboard data as the final output

Demo Video :

3. Documentation

Objective :

The objective of this project is to develop a Computer Vision system that automatically detects and extracts scoreboard information from a video. The system processes video frames, identifies the scoreboard region, extracts the relevant scoreboard area, and recognizes the displayed information to produce structured output.

System Architecture :

```
        INPUT VIDEO
           |
           ▼
       FRAME EXTRACTION
           |
           ▼
     SCOREBOARD DETECTION
          (YOLO)
           |
           ▼
     SCOREBOARD CROPPING
```

|

▼

IMAGE PREPROCESSING

|

▼

OCR / TEXT

EXTRACTION

|

▼

DATA PROCESSING

|

▼

STRUCTURED SCOREBOARD

OUTPUT

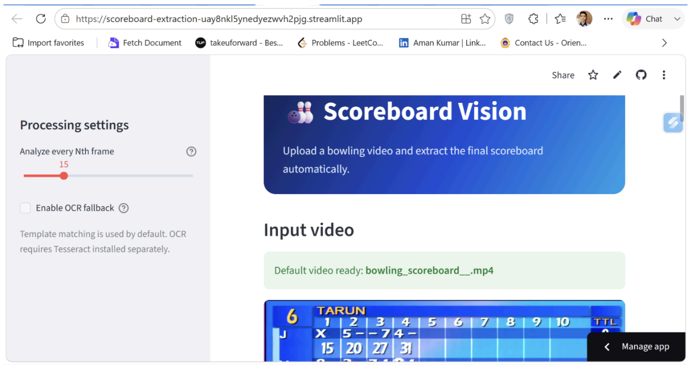
The proposed pipeline processes the input video frame-by-frame. A trained YOLO object detection model identifies the scoreboard region. The detected region is cropped and passed through the text extraction pipeline. The recognized information is then processed and presented as structured scoreboard data.

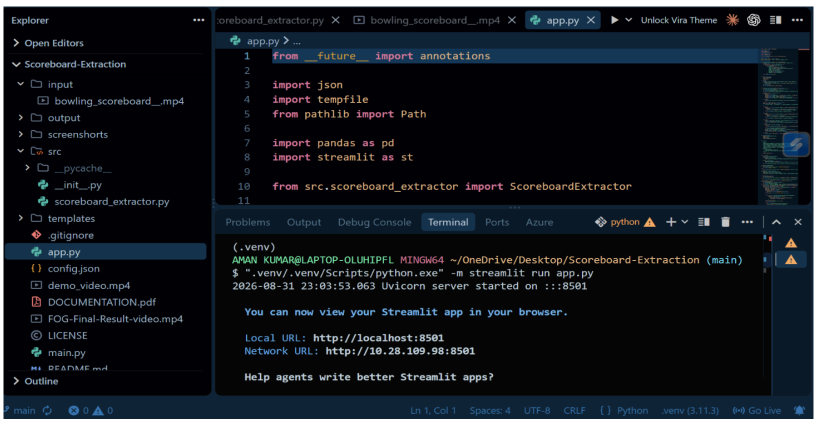
Technologies Used :

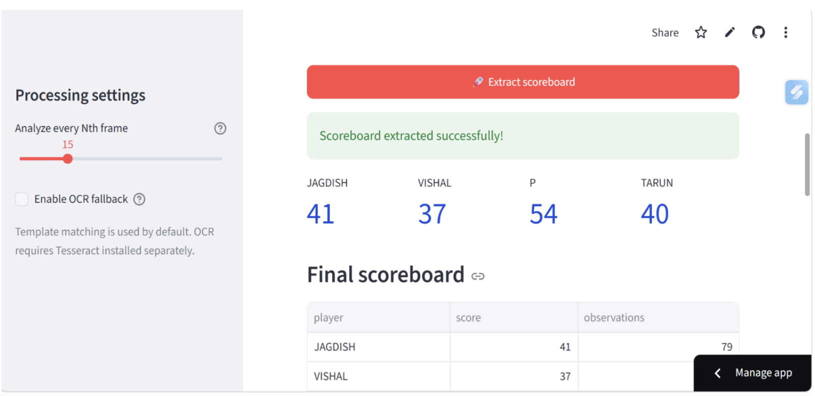
Technology	Purpose
Python	Core programming language
OpenCV	Video and image processing
YOLO	Scoreboard detection
OCR	Text/number extraction
Streamlit	User interface / deployment
NumPy	Numerical/image processing
Git & GitHub	Version control and source-code hosting

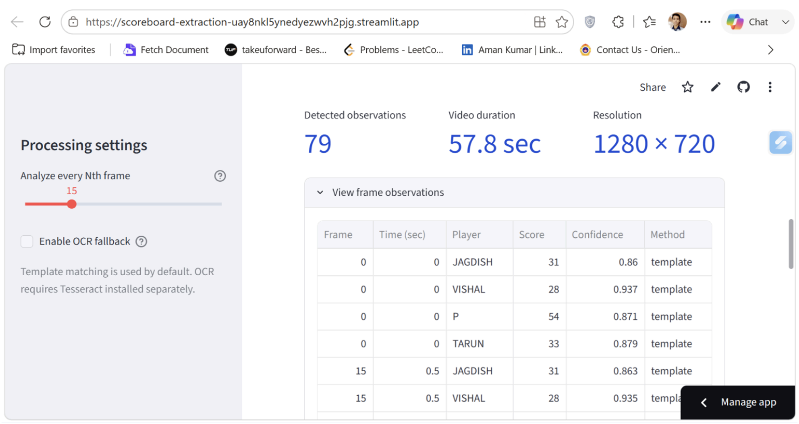
Working flow :

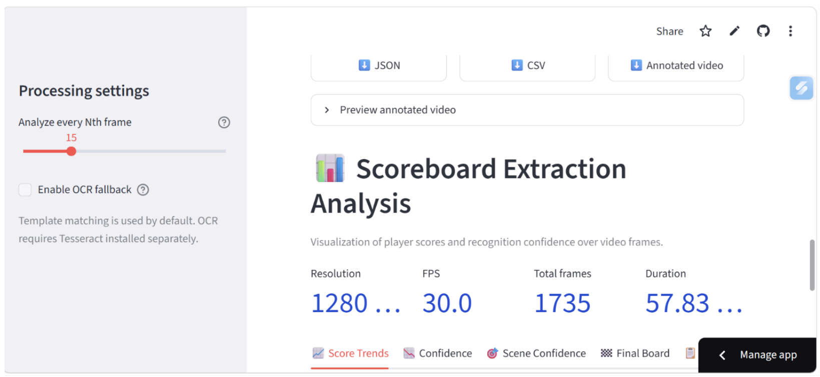
Screenshots:

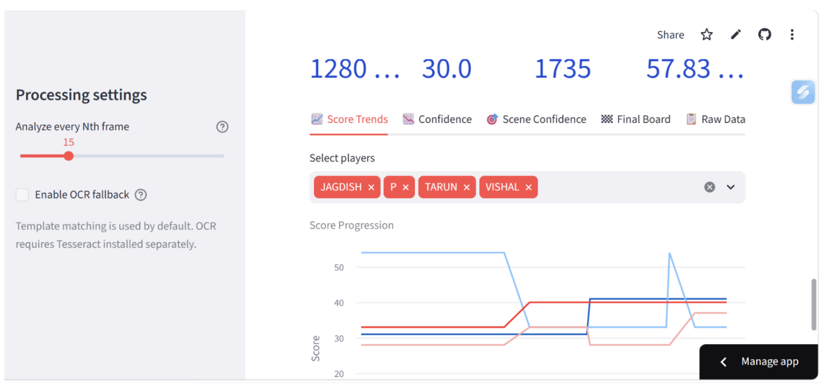

https://scoreboard-extraction-uay8nkl5ynedyezwvh2pjg.streamlit.app
Import favorites
Fetch Document
takeuforward - Bes...
Problems - LeetCo...
Aman Kumar | Link...
Contact Us - Orien...
Share
Processing settings
Analyze every Nth frame
15
Enable OCR fallback
Template matching is used by default. OCR requires Tesseract installed separately.
Scoreboard Vision
Upload a bowling video and extract the final scoreboard automatically.
Input video
Default video ready: bowling_scoreboard__.mp4
TARUN
Manage app


Explorer
Open Editors
Scoreboard-Extraction
input
bowling_scoreboard__.mp4
output
screenshorts
src
__pycache__
__init__.py
scoreboard_extractor.py
templates
.gitignore
app.py
config.json
demo_video.mp4
DOCUMENTATION.pdf
FOG-Final-Result-video.mp4
LICENSE
main.py
Outline
app.py
from __future__ import annotations
import json
import tempfile
from pathlib import Path
import pandas as pd
import streamlit as st
from src.scoreboard_extractor import ScoreboardExtractor
Problems
Output
Debug Console
Terminal
Ports
Azure
(.venv)
AMAN KUMAR@LAPTOP-OLUHIPFL MINGW64 ~/OneDrive/Desktop/Scoreboard-Extraction (main)
$ ".venv/.venv/Scripts/python.exe" -m streamlit run app.py
2026-08-31 23:03:53.063 Uvicorn server started on :::8501
You can now view your Streamlit app in your browser.
Local URL: http://localhost:8501
Network URL: http://10.28.109.98:8501
Help agents write better Streamlit apps?
main
Ln 1, Col 1
Spaces: 4
UTF-8
CRLF
Python
.venv (3.11.3)
Go Live


Share
Processing settings
Analyze every Nth frame
15
Enable OCR fallback
Template matching is used by default. OCR requires Tesseract installed separately.
Extract scoreboard
Scoreboard extracted successfully!
JAGDISH 41
VISHAL 37
P 54
TARUN 40
Final scoreboard
player
score
observations
JAGDISH
41
79
VISHAL
37
Manage app







Conclusion :

The developed system demonstrates an end-to-end Computer Vision pipeline for extracting scoreboard information from video. The system detects the scoreboard region, isolates the relevant area, extracts the displayed information, and presents the resulting data to the user. The modular design allows individual components such as detection, preprocessing, OCR, and post-processing to be further improved independently.